PRODUCT REQUIREMENT DOCUMENT (PRD)
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Project Name: OpsResolve: AI Conflict Resolver for Industrial Operations

1. Project Overview

- **Problem Statement:** In organisations with **multiple departments, conflicting decisions are common** — for example, Operations requesting budget increases while Finance enforces cost-cutting. **These conflicts are rarely detected in real time** and are often resolved through slow, manual escalation processes that rely on back-and-forth email chains and ad-hoc meetings. There is no structured system to capture, surface, and intelligently mediate these competing decisions. As a result, organisations face delayed projects, inefficient resource utilisation, and unresolved tensions between teams.

- **Target Domain:** Industrial Operations Coordination — including Production Scheduling, Maintenance Planning, Logistics & Resource Allocation, and Quality Control.

- **Proposed Solution Summary:**
  - OpsResolve is a role-based web platform designed to improve cross-department coordination in industrial operations. Departments **submit operational records through structured forms or AI-assisted input methods** such as document upload (PDF/image) and natural language descriptions.
  - The system **combines rule-based logic with AI-powered analysis to detect conflicts in real time**, such as overlapping schedules, shared resource clashes, or dependency issues. When a conflict is identified, the **system immediately alerts users and generates an AI-supported analysis**, including a summary of the conflict, its impact, and a recommended resolution with a confidence score.
  - Users can view and manage conflicts through an interactive dashboard. Supervisors can coordinate responses by notifying involved parties or scheduling discussions, while managers have full authority to accept AI recommendations or override them with justified decisions. All actions are recorded to maintain a transparent audit trail.
  - In addition, OpsResolve provides an **AI-powered machine request feature**, allowing users to **request equipment using voice or text input**, with the system automatically extracting requirements and suggesting available resources.

2. Background & Business Objective

- **Background of the Problem:**
  - Manufacturing and industrial operations have traditionally relied on siloed planning systems. Production uses one scheduling tool, Maintenance operates on a separate CMMS (Computerised Maintenance Management System), and Logistics manages deliveries independently. These systems rarely communicate in real time, leading to limited visibility across departments.

- As a result, conflicts often only surface at the point of execution — for example, when a machine is unexpectedly unavailable due to maintenance during a production shift, or when multiple departments unknowingly reserve the same equipment within overlapping time windows.
- The manual resolution process — involving supervisors, shift managers, and sometimes plant directors — is time-consuming and inefficient. It typically requires multiple rounds of communication and coordination, increasing the likelihood of delays and decisions that may resolve one issue while creating downstream conflicts.

- **Importance of Solving This Issue:**
  - **Reducing unplanned downtime:** Early detection of scheduling and resource conflicts prevents costly production interruptions and improves operational continuity.
  - **Improving resource utilisation:** Real-time visibility of shared assets such as machines and equipment helps avoid double-booking and ensures optimal allocation.
  - **Accelerating conflict resolution:** AI-supported analysis and recommendations significantly reduce the time required for managers to evaluate and resolve conflicts.
  - **Creating an operations audit trail:** Every conflict, recommendation, and decision is systematically recorded, enabling traceability and future optimisation of operational planning.
  - **Breaking down departmental silos:** A unified conflict management workspace enhances transparency and facilitates structured communication across departments.

- **Strategic Fit / Impact:**
  - OpsResolve **aligns with the broader industrial digitalisation trend (Industry 4.0)**, where operational intelligence is enhanced through AI-assisted systems. Rather than replacing existing tools, OpsResolve acts as a coordination layer that connects production scheduling, maintenance planning, and logistics operations.
  - The system leverages a combination of rule-based conflict detection and AI-powered analysis using **Google Gemini 2.5 Flash** to **interpret operational data, generate conflict insights, and recommend actionable resolutions**. This hybrid approach ensures both reliability in detection and flexibility in reasoning.
  - By integrating AI-assisted decision support into everyday operational workflows, OpsResolve enables organisations to move from reactive conflict handling to proactive coordination. The system is designed to be scalable and adaptable across

different industrial environments where multiple departments must coordinate shared resources and schedules.

3. Product Purpose

3.1. Main Goal of the System:

To **streamline cross-department coordination** by automatically detecting scheduling, resource, and workflow conflicts in real time, assessing their operational impact, and supporting fast, structured resolution through AI-assisted decision-making.

3.2. Intended Users (Target Audience):

- **Data Entry**
  Responsible for submitting operational records (e.g., production schedules, maintenance windows, logistics requests).
  - Access: Submit records, view conflicts (read-only)
- **Supervisor**
  Oversees departmental operations and coordinates conflict handling between teams.
  - Access: View conflicts, notify involved parties, schedule coordination meetings
- **Manager**
  Holds decision-making authority for conflict resolution.
  - Access: Full control — review AI analysis, accept recommendations, or override decisions with justification

4. System Functionalities

4.1. Description: OpsResolve operates as a structured conflict intelligence system for industrial operations coordination. When an operational record is submitted, the system immediately performs a **rule-based conflict detection check** against all active records across departments. Conflicts are identified based on overlapping time windows, shared resources (e.g., machines, equipment), or location and shift overlaps.

Once a conflict is detected, an AI engine (Google Gemini 2.5 Flash) **generates a structured analysis** that includes a plain-language explanation of the conflict, its operational impact, and a recommended resolution with a confidence score. This **allows users to quickly understand not just that a conflict exists, but why it occurs and how it can be resolved**.

The system provides real-time feedback through a conflict alert modal and updates the conflict dashboard instantly. **Supervisors can coordinate responses by notifying involved parties or scheduling discussions**, while **managers review the AI-generated insights and make final decisions by accepting or overriding recommendations**. All actions are recorded to ensure traceability and accountability.

4.2. Key Functionalities:

- **1. Entry & Role Selection:** Users select their role (Data Entry, Supervisor, Manager) and department upon entering the system. This determines their access permissions and visible actions within the platform.

- **2. New Operational Record Submission:** Users can submit operational records through:
  - Structured form input
  - AI-assisted document upload (PDF/Image) — automatically extracts relevant fields
  - Natural language input (for certain features)

  Each record includes:
  - Record type (Production Schedule / Maintenance Window / Logistics Request / Quality Control Hold)
  - Title and description
  - Time window (start/end)
  - Resource details (machine IDs, equipment, location, shift)

  The system also supports:
  - **AI Fill:** Automatically populates fields from uploaded documents
  - **AI Imputation:** Suggests missing values based on historical records

- **3. Active Conflicts Dashboard:** Upon submission, the system immediately checks for conflicts using predefined logic:
  - **Cross-department:** same equipment OR same location + same shift + same date
  - **Same-department**: same equipment + same shift + same date

  If a conflict is detected:
  - A **Conflict Detected modal** appears instantly
  - Users can directly view conflict details and AI-generated insights

- **4. Decision Review & Audit Log:** A real-time dashboard displays all detected conflicts with:
  - Departments involved
  - Conflict type and severity
  - AI confidence score
  - Current status (Open / Coordinated / Resolved)

  Features include:

- Department-specific views
- Sorting by severity
- Show/hide resolved conflicts
- Instant updates when actions are taken

○ **5. Conflict Details & Coordination Actions**

Each conflict can be opened in a detailed view showing:

- Conflicting records and key attributes
- AI-generated analysis (cause, risk, impact)
- Recommended resolution with confidence score

Supervisors and managers can:

- Notify all involved parties
- Schedule coordination meetings

○ **6. Decision Review & Resolution (Manager Only)**

Managers have full authority to resolve conflicts by:

- Accepting the AI recommendation
- Overriding the recommendation with a justified note

Once resolved:

- Conflict status updates immediately
- Record is removed from active conflicts

○ **7. Audit Log & Traceability**

All conflict resolutions are permanently recorded, including:

- Decision type (accepted or overridden)
- Manager's note
- AI recommendation presented
- Timestamp and user identity

This ensures transparency and supports future operational analysis.

○ **8. AI-Powered Machine Request Module**

Users can request machines using voice or text input (e.g., "I need a welding machine tomorrow morning"). The system:

- Extracts intent using AI
- Identifies required machine type, date, and shift
- Suggests available machines
- Allows users to confirm and submit bookings

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

OpsResolve uses **Google Gemini 2.5 Flash** as its **AI reasoning engine** to support conflict analysis, field extraction, and decision recommendation. The model is integrated into a simulated multi-step workflow, where different reasoning tasks are handled through structured prompt calls rather than a single monolithic request.

Gemini was selected for the following capabilities relevant to industrial operations:

- **Natural language understanding for operational inputs:** Gemini effectively interprets unstructured inputs such as user descriptions, voice/text requests, and uploaded work orders (PDF/images), enabling the system to extract key fields such as time windows, equipment, and task details.
- **Context-aware conflict explanation:** The model generates clear, human-readable explanations of why conflicts occur, translating technical scheduling overlaps into understandable insights for users across different roles.
- **Structured JSON output generation:** Gemini reliably produces structured, schema-based outputs when guided by prompt design, allowing the system to consistently display conflict summaries, severity indicators, recommendations, and confidence scores in the UI.
- **Flexible reasoning across multiple tasks:** Instead of relying on a single AI call, OpsResolve uses Gemini in multiple stages — including field extraction, conflict interpretation, and recommendation generation — enabling modular and reusable AI logic across the system.
- **Domain-adaptable prompting:** Through carefully designed prompts, the system incorporates industrial terminology (e.g., machine IDs, shifts, maintenance windows), ensuring outputs remain relevant to real-world operational scenarios.
- **Real-time performance suitability:** Gemini 2.5 Flash provides fast response times suitable for interactive applications, allowing conflict analysis and recommendations to be generated within seconds after record submission.

4.3.2. Prompting Strategy:

OpsResolve implements a simulated multi-agent workflow using Google Gemini 2.5 Flash, where **different reasoning tasks are handled through a sequence of structured prompt**

calls. Instead of relying on a single AI response, the system decomposes the problem into multiple stages, each responsible for a specific function in the conflict analysis pipeline.

This approach improves clarity, modularity, and consistency of outputs while maintaining real-time performance.

Agent Roles & Responsibilities

Although implemented through sequential prompt calls, the system is conceptually structured into five specialised agents:

Agent	Role & Output
Coordinator (Workflow Controller)	Acts as the central controller of the workflow. It **manages the sequence of AI calls**, passes outputs between stages, and determines when the process is complete. It also **handles system actions** such as triggering notifications and escalating conflicts for manager review.
Input Processing Agent	Processes raw user input, which may come from: ● Structured forms ● Natural language descriptions ● Uploaded documents (PDF/Image via AI extraction) It extracts and normalises key fields: ● Record type ● Time window ● Resource identifiers (machine, location, shift) ● Department context If required fields are missing or unclear, the system flags them for user clarification before proceeding.

Conflict Detection Layer (Rule-Based + AI Support)	Unlike a fully AI-driven approach, OpsResolve uses **deterministic rule-based logic** to detect conflicts: ● Same equipment ● Same location + shift + date ● Cross-department overlaps AI is then used to: ● Interpret the nature of the conflict ● Identify conflicting attributes ● Provide a structured explanation
Impact Analysis Agent	Evaluates the operational impact of the detected conflict. It generates: ● A plain-language summary of consequences ● Risk and delay implications ● Severity level (Low / Medium / High / Critical) This helps prioritise conflicts and supports decision-making.
Decision Agent	Generates a recommended resolution based on: ● Conflict type ● Operational constraints ● Impact severity Outputs include: ● Suggested action (e.g., reschedule, reallocate resource) ● Affected parties ● Justification (rationale) ● Confidence score

- **Workflow Execution**
  The workflow follows a structured sequence:
  Input → Detection → Analysis → Recommendation

  Each stage receives the output of the previous stage, ensuring that:
  - Context is preserved
  - Reasoning is consistent

- Outputs are explainable

Although described as "agents," this architecture is implemented as a **controlled sequence of AI prompt calls within the backend (runAgentLoop)** rather than independent autonomous agents.

4.3.3. Context & Input Handling:

OpsResolve uses a **structured context object** assembled by the backend and passed into each AI prompt call. This ensures that the model does not rely solely on raw text input, but instead receives relevant system data such as active records, resources, and prior processing outputs. This approach improves consistency, reduces ambiguity, and enables more accurate conflict analysis.

The context object includes the following components:

Field	Purpose
departments	List of active departments and their associated records
resources	Resource data including machine IDs, equipment, and availability status
schedules	Active time-window records relevant to the current request
current_request	The normalised record being evaluated
workflow_state	Current processing state (e.g., UNDER_ANALYSIS, AWAITING_DECISION, RESOLVED)
prior_outputs	Outputs generated from previous processing steps

Rather than using full tool-calling mechanisms, the **backend retrieves required data** (e.g., schedules, resource availability) and **injects it directly into the prompt context before sending it to the AI model.**

Handling unstructured inputs:

Operational records are often submitted in unstructured formats, such as free-text descriptions, voice input, or uploaded documents. These inputs may contain incomplete information, informal language, or domain-specific shorthand (e.g., "M07 down Thurs AM").

To handle this, OpsResolve applies AI-assisted preprocessing:

- **Field extraction:**
  The AI model extracts structured fields such as record type, time window, resource identifiers, and department context from unstructured input sources.
- **Multi-source input support:**
  Inputs can come from:
  - Manual form entry
  - Natural language text or voice
  - Uploaded documents (PDF/image via AI extraction)
- **Ambiguity handling:**
  If critical fields cannot be reliably extracted, the system flags the issue and prompts the user for clarification instead of making assumptions.
- **Missing data control:**
  Records with missing mandatory fields (e.g., time window, resource identifiers) are not processed for conflict detection until the required information is completed.
- **Confidence scoring:**
  Extracted fields are associated with a confidence level. Fields with low confidence may require user confirmation before submission is finalised.

Handling Large and Complex Inputs:

To ensure performance and reliability, the system applies constraints to input size and complexity:

- Extremely long descriptions are truncated to maintain efficient processing
- Large resource lists are summarised to reduce prompt size
- Inputs exceeding defined limits are rejected with clear validation messages

These safeguards ensure that the AI model operates within practical limits while maintaining transparency to the user.

4.3.4. Stateful Decision-Making:

OpsResolve maintains a **persistent workflow state** for each conflict throughout its lifecycle. This state is stored in the backend database (Firebase Firestore) and is passed between processing steps to ensure continuity, traceability, and consistent decision-making.

- **Agent continuity**:
  Although implemented as a sequence of AI prompt calls, each stage of the workflow receives the accumulated outputs from previous stages. For example, the recommendation generation step has access to:
  - The normalised input record
  - Detected conflict details
  - Impact analysis and severity

  This ensures that decisions are made with full awareness of prior processing results and context.

- **Conflict lifecycle tracking**:
  Each conflict progresses through a clearly defined set of states:
  DETECTED → UNDER_ANALYSIS → AWAITING_DECISION → RESOLVED
  - **DETECTED:** Conflict identified by rule-based detection
  - **UNDER_ANALYSIS:** AI analysis is being generated
  - **AWAITING_DECISION:** Conflict is ready for manager review
  - **RESOLVED:** Final decision has been applied

  These states are stored and updated in the database, enabling real-time UI updates and clear tracking of conflict progress.

- **Manager decision context**:
  When a manager opens a conflict in the Decision Review interface, the system retrieves the full context, including:
  - Original records involved
  - AI-generated analysis (cause, impact)
  - Recommended resolution with confidence score

  This allows managers to make informed decisions based on both system data and AI insights, rather than relying on fragmented information.

4.3.5 Generation of Structured, Actionable Outputs

OpsResolve ensures that all AI-generated results are returned in a **structured JSON format**. This guarantees that outputs are machine-readable, consistently displayable in the user interface, and suitable for audit and tracking purposes.

Rather than relying on unstructured text responses, the system enforces a predefined output schema for every AI processing stage. This allows seamless integration between backend processing and frontend visualisation.

The final combined output structure is as follows:

Output Field	**Source & Purpose**
record_type	Classified record type extracted by Input Agent (Production Schedule / Maintenance Window / Logistics Request / Quality Control Hold)
conflict_type	Conflict classification based on detection logic (Time / Resource / Combined conditions)
conflicting_fields	Specific attributes causing the conflict (e.g., overlapping time windows, shared resource IDs)
impact_summary	Plain-language explanation of the operational impact generated by AI analysis
severity	Severity level indicating urgency (Low / Medium / High / Critical)
suggested_action	Recommended resolution (e.g., rescheduling, resource reallocation)
affected_parties	Departments and resources impacted by the conflict and proposed solution
rationale	Short justification explaining why the suggested action minimises disruption

confidence	Confidence score (0–1) indicating reliability of the AI-generated outputs.

4.3.6 Fallback & Failure Behaviour

OpsResolve is designed to handle real-world uncertainties by **providing robust fallback mechanisms** for three main scenarios: **AI model failures, ambiguous inputs, and missing data**. Each scenario follows a defined handling strategy to ensure the system remains usable and reliable.

AI model failures:

1. **Response validation**: Each AI output is checked to ensure it conforms to the expected structured format (JSON schema). Invalid or incomplete responses are identified before being used in the system.
2. **Retry mechanism**: If validation fails, the system retries the AI call once with stricter prompt instructions to enforce the required output format.
3. **Graceful degradation**: If the retry still fails, the system continues without AI-generated analysis. The conflict is displayed with an **"AI analysis unavailable"** notice, allowing users to proceed with manual decision-making.
4. **Error logging**: Failed AI responses are logged along with relevant input data for debugging, monitoring, and future prompt improvements.

Ambiguous and missing data:

Ambiguous and Missing Data:

5. **Input validation control**: Records missing critical fields (e.g., time window, resource identifiers) are not processed for conflict detection until required information is provided.
6. **Targeted clarification prompts**: Instead of rejecting the entire submission, the system prompts users to supply only the missing or unclear information, improving usability and efficiency.
7. **Partial processing with caution:** If only non-critical fields are ambiguous, the system proceeds with conflict detection using imputed data, while highlighting uncertainty in the output (mentioned AI suggested) and reducing confidence scores.

5. User Stories & Use Cases

5.1 User Stories

ID	User Story

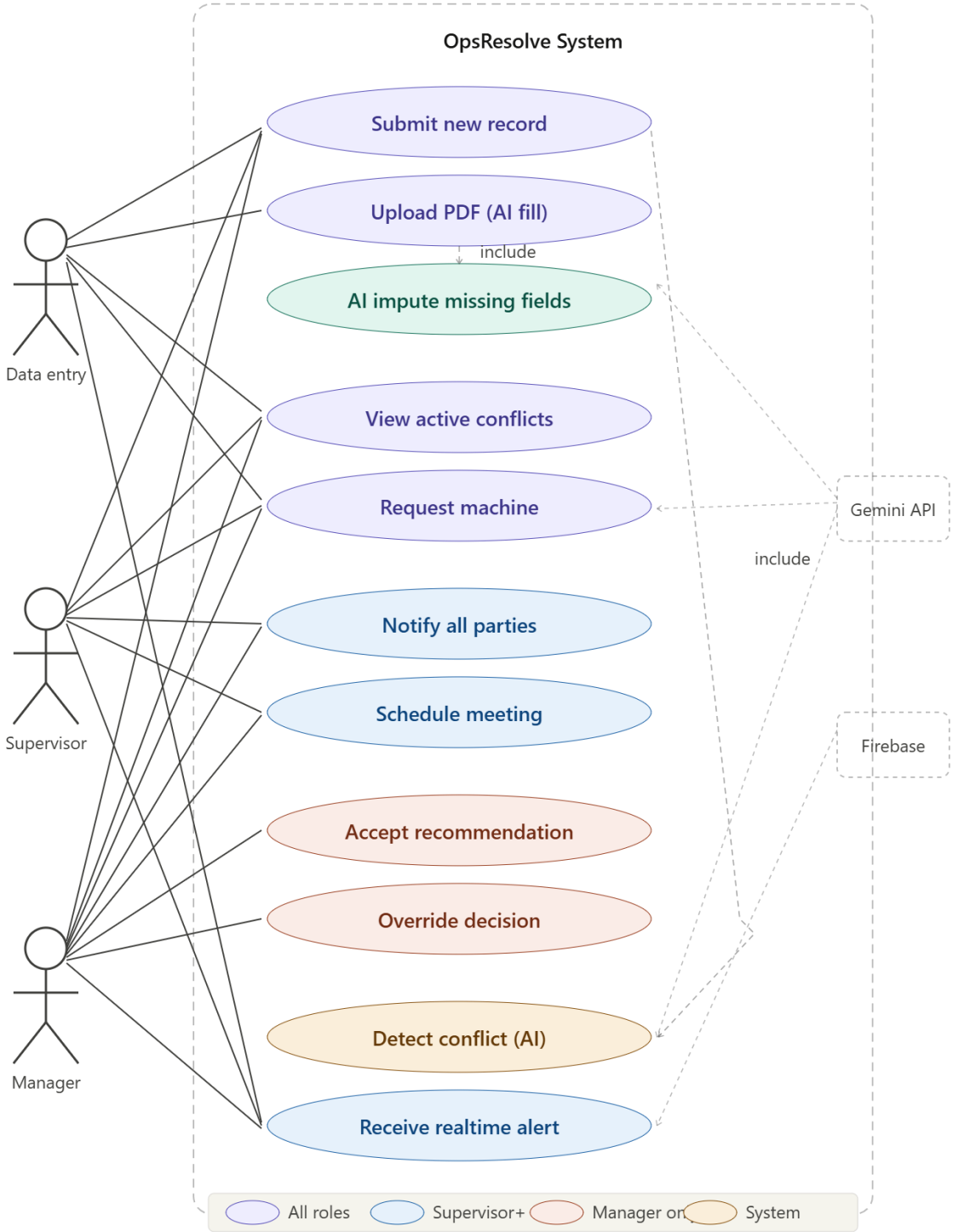
US-01	As a production planner, I want to submit my production schedule so that the system can automatically detect conflicts with existing maintenance and resource allocations before execution.
US-02	As a maintenance supervisor, I want to be notified immediately when my planned maintenance conflicts with other operations so that I can proactively adjust the schedule.
US-03	As a logistics coordinator, I want to identify shared equipment conflicts across departments so that I can prevent operational delays before tasks begin.
US-04	As an operations manager, I want to view a prioritised list of active conflicts sorted by severity and confidence so that I can focus on the most critical issues first.
US-05	As a manager, I want to open a conflict and see AI-generated analysis, impact, and recommended actions so that I can make informed decisions quickly.
US-06	As a manager, I want to accept or override AI recommendations with a justification so that decisions remain accountable and transparent.
US-07	As a user, I want to access an audit log of past conflicts and resolutions so that I can review historical decisions and improve future planning.

5.2 Use Cases (Main Interactions)

- **Prompt to UI Flow:**
  - User uploads a PDF or speaks a request
  - Gemini parses the content

- Fields are extracted and mapped to form components
- The New Record form auto-fills with the extracted + imputed values
- User reviews and submits
- Conflict detection runs
- UI renders either a success banner or a Conflict Detected modal.

- **Flow generation:**
  - After submit, the system determines the next screen based on the result: no conflict shows the clean form ready for next entry; conflict detected shows the ConflictDetectedModal with AI recommendation
  - Clicking "View Conflict Details" renders the full ConflictPopup with record cards, AI analysis, and action buttons.



6. Features Included (Scope Definition)

- **Role & Department-Based Access Control**
  Entry page allowing users to select their role (Data Entry, Supervisor, Manager) and department, which determines their access permissions and available actions within the system.

- **AI-Assisted Operational Record Submission**
  Structured form supporting four record types:
  - Production Schedule
  - Maintenance Window
  - Logistics Request
  - Quality Control Hold

  Includes:
  - AI Fill: Automatic extraction of fields from uploaded documents (PDF/image)
  - AI Imputation: Suggestion of missing values based on historical records

- **Real-Time Conflict Detection Engine**
  Hybrid detection using rule-based logic:
  - Same equipment conflicts
  - Same location + shift + date conflicts
  - Cross-department overlaps

  Conflicts are detected instantly upon record submission.

- **Conflict Detected Alert (Real-Time Modal)**
  Immediate visual alert displayed when a conflict is identified, allowing users to access conflict details without navigating away.

- **Active Conflicts Dashboard**
  Real-time dashboard displaying:
  - Conflicts relevant to the user's department
  - Severity level and status
  - AI confidence score
  - Filtering and sorting options

  Includes live updates when actions are taken.

- **Conflict Details & AI Analysis View**
  Detailed conflict interface showing:
  - Conflicting records and attributes
  - AI-generated explanation (cause, impact, risk)
  - Recommended resolution with confidence score

- **Manager Decision Panel**
  Managers can:
  - Accept AI recommendation
  - Override recommendation with mandatory justification

  Decisions update system state instantly.

- **AI-Powered Machine Request Module**
  Users can request equipment using voice or text input. The system:
  - Extracts intent (machine type, date, shift)
  - Suggests available resources
  - Allows confirmation and booking

7. Features Not Included (Scope Control)

- Direct integration with existing CMMS, ERP, or production scheduling software (e.g., SAP, Oracle, Maximo). OpsResolve is a standalone coordination layer in v1.
- Automated conflict resolution without manager approval. All resolutions require human sign-off.
- Predictive conflict forecasting (proactive identification of future conflicts before records are submitted).
- Real-time machine telemetry or IoT sensor integration.
- Multi-site operations across different plant locations.
- Mobile application (iOS/Android). The system is a responsive web application only.
- Advanced analytics, OEE impact reporting, or conflict pattern visualisation dashboards.
- Multi-user simultaneous editing of the same conflict record.

8. Assumptions & Constraints

- **LLM Cost Constraint:**
  OpsResolve uses Google Gemini 2.5 Flash for AI-assisted processing. The estimated token usage per user session is approximately 3,000–5,000 tokens, based on a multi-step prompting approach (input processing, analysis, and recommendation generation).

  In an operational environment with 10–20 conflict events per shift, this results in an estimated 30,000–100,000 tokens per shift.

  To manage inference cost efficiently, the system incorporates the following design optimisations:

- **AI result caching:** Previously resolved conflict patterns are stored in the database. Similar future conflicts can reuse cached outputs instead of triggering new AI calls.
- **Input compression:** Large resource lists are summarised before being passed into prompts, reducing token usage for resource-heavy records.
- **Lazy AI invocation:** AI analysis is only triggered when a conflict is opened in the detailed view, rather than during initial detection. This avoids unnecessary AI calls for low-priority or quickly resolved conflicts.
- **Prompt segmentation:** Instead of a single large prompt, the system uses multiple smaller, focused prompts for different stages (extraction, analysis, recommendation), improving efficiency and control.

- **Technical Constraints:**
  - The frontend (React + Vite) is a **prototype implementation** and is not hardened for production deployment without security review.
  - The backend (Node.js + Express) does not include full authentication middleware in v1. Role selection is **trust-based** for the demo environment.
  - Firebase Firestore is used for persistence. This limits complex relational querying compared to traditional relational databases.
  - Conflict detection is processed **synchronously upon submission**. At higher loads (e.g., 20+ concurrent submissions), a queue-based architecture would be required, which is not included in the current scope.

- **Performance Constraints:**
  - Target AI response time: **< 10 seconds per conflict analysis**
  - Timeout threshold: **20 seconds**, after which the system falls back to manual handling
  - Conflict detection time: **< 2 seconds per submission** (up to ~500 active records)
  - The demo system supports approximately **up to 50 concurrent users**

- **User Input:**
  - Users are assumed to provide **accurate and valid operational data**. The system does not include advanced validation or fraud detection mechanisms.
  - Time inputs follow a **24-hour format** and are assumed to be within a **single local timezone** (no multi-timezone support in v1).
  - The system assumes **one decision-making authority (Manager)** per operational environment.
  - A minimum input length is required for meaningful AI processing. Extremely short or incomplete descriptions are rejected to maintain output quality.

9. Risks & Questions Throughout Development

- **Design Similarity:**
  - **AI Resolution Similarity:** How are we ensuring the AI does not generate identical resolution recommendations for every conflict regardless of context, producing generic outputs that do not reflect the actual operational differences between departments?
- **Frame Stability:**
  - **Conflict Record Integrity:** Does data entry user need to manually correct or resubmit a record if the Input Agent misclassifies the record type after submission?
- **Drawing Interpretation**:
  - **Unstructured Input Misinterpretation:** What will happen if the Input Agent is unable to extract a required field from a vague or informal description submitted by a Data Entry user (e.g. "M07 down Thurs AM")?